Pre-Analysis Plan

The following pre-analysis plan describes the data and empirical setup for investigating the following hypotheses:

Hypothesis 1)	Having daughters leads to stronger support for gender equality (as measured by attitudes).
Hypothesis 2)	Having daughters leads to more left-leaning political preferences.
Hypothesis 3)	Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)
Hypothesis 4)	Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

While testing all hypotheses, we rely on the same data source and sample, key independent variable, and control variables, described in the following.

1. Data and Sample

We use panel data from the German Socio-Economic Panel (SOEP). We exclude individuals with missing responses on either of the dependent variables, on the key independent variable related to children's gender composition, or on control variables. We include all survey years in which the outcome variables are available (the precise window depends on the fielding of relevant items, typically between 2002 and 2022).

2. Key Independent Variable

Our key independent variable of interest is a continuous count variable, $daughters_nr$, which captures the total number of daughters a respondent has. We extract this information from the variable **kidsex** in the SOEP dataset. This allows us to estimate the marginal effect of each additional daughter on support for gender equality.

Note that we do not employ categorical indicators for specific daughter counts, as our primary interest lies in identifying a consistent directional effect across increasing numbers of daughters rather than detecting threshold or nonlinear effects. Additionally, focusing on a single continuous variable simplifies interpretation and reduces the risk of overfitting in subsamples, especially when comparing effects across respondent gender groups in Hypothesis 4. And finally, this strategy is also in line with previous research in this domain (Oswald & Powdthavee, 2010; Washington, 2008).

3. Control Variables

All models will incorporate a set of control variables designed to account for potential factors that may influence the respondents' attitudes. Covariates are selected to capture core demographic, socioeconomic, and familial characteristics that are plausibly correlated with both the treatment and the outcomes of interest. Based on the variable descriptions available at paneldata.org, we insured to choose variables that do not restrict the sample size when testing the models for the four hypotheses. The following table lists our selected controls, including the raw number of valid observations per variable (N).

Symbol	Variable Description	Source	N
d_male	Dummy variable, 1 if respondent is male, 0 if female	sex	
ln_age	Natural logarithm of respondent's age	bioage	
ln_net_income	Natural logarithm of monthly net household income	pglabnet	487 938
$d_married$	Dummy, 1 if currently married	pgfamstd = 1	812 381
$d_divorced$	Dummy, 1 if currently divorced	pgfamstd = 4	812 381
$d_separated$	Dummy, 1 if currently separated	pgfamstd = 2	812 381
$d_vocational$	Dummy, 1 if respondent has completed vocational education	pgcasmin > 3 & pgcasmin <= 7	794 006
d_higher_edu	Dummy, 1 if respondent has completed a university degree	pgcasmin >= 8 & pgcasmin <= 9	794 006
$d_migback$	Dummy, 1 if respondent has migration background	migback == 2 & migback == 3	1 446 266

The selection and operationalization of control variables are grounded in prior literature. Gender, age, and income are included as standard covariates, consistent with early studies such as Wilcox (1990). Marital status and educational attainment are represented by a set of dummy variables for each category, which we consider the most precise specification for capturing their potential effects (Oswald & Powdthavee, 2010). Migration background is incorporated as a control in line with recent contributions emphasizing its relevance for political preferences (Rehm et al., 2022). By contrast, we refrain from controlling for religion, as in Washington (2008) and Wilcox (1990), because doing so would substantially reduce the sample size and thereby undermine the statistical power of our analyses. To isolate the effect of having an additional daughter from the effect of having an additional child, we include fixed effects for the total number of children. This approach ensures that our estimates capture the impact of a child's gender rather than confounding it with family size effects (Washington, 2008).

In the next section, we describe the dependent variables and models as well as how we interpret the outcomes for each hypothesis separately.

4. Operationalization and Testing of Hypotheses

4.1. Hypothesis 1) Having daughters leads to stronger support for gender equality, as measured by attitudes.

Dependent Variable

There is no survey item in the SOEP dataset specifically asking for the of support for gender equality. Hence, we rely on a dependent variable that proxies this attitudinal dimension. It is based on the SOEP item **plh0174**, in which respondents are asked to state their agreement with their satisfaction with housework. This variable has 577 087 valid observations in the core dataset. We construct a binary variable, $\boldsymbol{satis_housew}$, coded as 1 if the respondent selects a strong satisfaction, and 0 otherwise. A value of 1 reflects high satisfaction with the current division of housework. We interpret this as a proxy for successful negotiation of domestic responsibilities and, by implication, greater gender equality in the household. This assumes that satisfaction arises from balanced contributions by both partners and not from acquiescence to traditional roles. Conversely, a value of 0 indicates dissatisfaction and may signal perceived inequity in the domestic division of labor.

Model

To test whether having daughters is linked to greater support for gender equality attitudes, we use panel data with total number of children fixed-effects to control for the additional effect of having daughters over having children in general. We estimate a logistic regression, since our outcome variable is binary, with standard errors clustered by the respondent. The model estimates the likelihood that respondents support gender equality based on how many daughters they have, while controlling for other characteristics and differences across survey years. Specifically, the specification is as follows:

$$logit(\Pr(\boldsymbol{satis_housew}_{it} = 1)) = \beta_0 + \beta_1 daughters_nr_{it} + X'_{it}\gamma + e_{it} \qquad (1)$$

with $\boldsymbol{satis_housew}_{it}$ being the dependent variable observed for respondent i at time t and X_{it} being a matrix of control variables. The term β_0 denotes a constant, β_1 and γ_k refer to the regression coefficients of the key independent and k-th control variable, respectively, and, finally, e_{it} is the residual.

Interpretation of Results

The primary coefficient of interest is that on $\boldsymbol{daughters_nr}$. A positive and statistically significant coefficient suggests that having more daughters is associated with greater support for gender equality. Statistical significance will be evaluated at the 10%, 5%, and 1% levels, with z-statistics also reported. If coefficients are negative or near zero, we will interpret this as a

lack of support for the hypothesis. Residual degrees of freedom are not reported in this analysis because logistic regressions rely on maximum likelihood estimation, which uses asymptotic inference rather than finite-sample variance partitioning.

4.2. Hypothesis 2) Having daughters leads to more left-leaning political preferences.

Dependent Variable

Political ideology is measured using the SOEP item **plh0004**, which asks respondents to place themselves on an 11-point left–right political scale from 0 ("Far left") to 10 ("Far right"). This variable has 90 972 valid observations in the core dataset. We recode this variable into a binary indicator $\boldsymbol{d_left}$, which equals one if the respondent reports a value between 0 and 3 (inclusive), corresponding to left-leaning political preferences, zero otherwise. This binary coding allows us to identify individuals who self-identify with the political left in a conceptually clear and replicable manner.

We do not use the full 11-point scale because the scale is highly subjective: respondents may differ in how they interpret and use intermediate points, making fine-grained comparisons difficult to interpret. By collapsing the scale into a binary indicator, we reduce this idiosyncratic measurement noise and focus instead on a meaningful ideological cluster — those who explicitly place themselves on the political left. This approach aligns with prior work (Oswald & Powdthavee, 2010) as well as our interest in detecting broader attitudinal shifts rather than capturing small, potentially arbitrary distinctions across the scale.

Model

We again use a logistic regression of the following form to test the hypothesis:

$$logit(\Pr(\boldsymbol{d_left}_{it} = 1)) = \beta_0 + \beta_1 daughters_nr_{it} + X'_{it}\gamma + e_{it} \quad (2)$$

with $\boldsymbol{d_left}_{it}$ being the dependent variable observed for respondent i at time t and X_{it} being a matrix of control variables. The term β_0 denotes a constant, β_1 and γ_k refer to the regression coefficients of the key independent and k-th control variable, respectively, and, finally, e_{it} is the residual.

We again control for total number of children fixed-effects to account for the additional effect of having daughters over having children in general. Standard errors are clustered by the respondent. These models predict the likelihood that a respondent has left-leaning political preferences based on how many daughters they have, while controlling for other relevant characteristics and differences across survey years.

Interpretation of Results

Again, the primary coefficient of interest is that on $daughters_nr$. A positive and statistically significant coefficient suggests that having more daughters is associated with more left-lean-

ing political preferences. Statistical significance will be evaluated at the 10%, 5%, and 1% levels, with z-statistics also reported. If coefficients are negative or near zero, we will interpret this as a lack of support for the hypothesis. Residual degrees of freedom are not reported in this analysis because logistic regressions rely on maximum likelihood estimation, which uses asymptotic inference rather than finite-sample variance partitioning.

4.3. Hypothesis 3) Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)

Dependent Variable

To measure gender equality in the household, we construct a continuous outcome variable that captures within-household gender inequality as measures by the net income gap. This metric is inspired by the intra-couple wealth gap analyzed by Rehm et al. (2022). Specifically, we calculate the difference between the respondent's individual net labor income measured in EUR (**pglabnet** with 487 938 valid observations) and the partner's labor net income retrieved via merge on partner ID (**parid**):

$$\boldsymbol{income_gap} = income_{own} - income_{partner}$$

This procedure limits the dataset to two persons with corresponding partner IDs, and hence implicitly removes singles from the analysis. We apply a transformation to normalize the distribution of the $\boldsymbol{income_gap}$ variable, which is skewed. A log transformation is commonly used for this purpose, but it is undefined for zero and negative values — which may occur in our case. To address this, we use the inverse hyperbolic sine (HIS) transformation, which behaves similarly to the logarithm for large values but remains defined for zero and negative inputs (Rehm et al., 2022). This allows us to retain the full range of observations without arbitrary truncation or recoding. Thus, our final variable is defined as follows:

$$\boldsymbol{ihs_income_gap} = \operatorname{asinh}(income_{own} - income_{partner})$$

This measure offers a complementary view of gender equality in the household as it captures relative economic contribution and bargaining power within couples.

Model

We estimate panel data models to assess whether having daughters influences the outcome of interest. To isolate the effect of daughters from the general effect of parenthood, we include fixed effects for the total number of children. This allows us to compare individuals with the same number of children but different gender compositions. Since our dependent variable is continuous, we estimate a linear model using ordinary least squares:

$$\boldsymbol{ihs_income_gap}_{it} = \beta_0 + \beta_1 daughters_nr_{it} + X'_{it}\gamma + e_{it} \qquad (3)$$

with X_{it} denoting a matrix of control variables. The term β_0 denotes a constant, β_1 and γ_k

refer to the regression coefficients of the key independent and k-th control variable, respectively, and, finally, e_{it} is the residual. We cluster standard errors at the household level to account for autocorrelation within respondents in the same household over time.

Interpretation of Results

Again, the primary coefficient of interest is that on $daughters_nr$. A negative and statistically significant coefficient suggests that having more daughters is associated with more gender equality in the household Statistical significance will be evaluated at the 10%, 5%, and 1% levels, with t-statistics also reported. If coefficients are positive or near zero, we will interpret this as a lack of support for the hypothesis.

4.4. Hypothesis 4) Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Dependent Variable

As for hypothesis 1, we rely on SOEP item **plh0174**, in which respondents are asked to state their satisfaction with housework. We construct a binary variable, $\boldsymbol{satis_housew}$, coded as 1 if the respondent selects a strong satisfaction, and 0 otherwise. A value of 1 reflects high satisfaction with the division of housework and is interpreted as an attitudinal proxy for perceived fairness in domestic responsibilities, assuming this satisfaction stems from balanced rather than traditional role allocations.

Model

We estimate a logistic regression to test the effect of having daughters on support for gender equality attitudes, allowing for gender-specific heterogeneity in the response. We include a continuous measure of the number of daughters and an interaction term with the respondent's gender. It tests whether having more daughters is associated with greater support for gender equality and whether that effect differs between men and women:

$$logit(\Pr(\boldsymbol{satis_houesew}_{it} = 1)) = \beta_0 + \beta_1 daughters_nr_{it} + \cdots \\ \beta_2 d_male_i + \beta_3 (daughters_nr_{it} \cdot d_male_i) + X'_{it}\gamma + e_{it} \quad (4)$$

For both specifications, $\boldsymbol{satis_housew}_{it}$ is the dependent variable observed for respondent i at time t and X_{it} denotes a matrix of control variables. The term β_0 denotes a constant, β_j and γ_k refer to the regression coefficients of the j-th key independent and k-th control variable, respectively, and, finally, e_{it} is the residual. As for the other hypotheses, we use panel data to estimate the model, include total number of children fixed-effects and cluster the standard errors at the respondents-level.

Interpretation of Results

The primary coefficients of interest is that on the interaction term $(daughters_nr_{it} \cdot d_male_i)$ because it captures whether and how the effect of having daughters on gender attitudes differs between men and women. A positive interaction coefficient supports the hypothesis that having daughters, especially among men, increases support for gender equality attitudes. Statistical significance will be evaluated at the 10%, 5%, and 1% levels, with z-statistics also reported. If coefficients are negative or near zero, we will interpret this as a lack of support for the hypothesis. Residual degrees of freedom are not reported in this analysis because logistic regressions rely on maximum likelihood estimation, which uses asymptotic inference rather than finite-sample variance partitioning.

References

Oswald, A. J., & Powdthavee, N. (2010). Daughters and left-wing voting. *The Review of Economics and Statistics*, 92(2), 213–227. https://doi.org/10.1162/rest.2010.11436.

Rehm, M., Schneebaum, A., & Schuster, B. (2022). Intra-couple wealth inequality: What's socio-demographics got to do with it? European Journal of Population, 38(4), 681–720. https://doi.org/10.1007/s10680-022-09633-4.

Washington, E. L. (2008). Female socialization: How daughters affect their legislator fathers' voting on women's issues. *American Economic Review*, 98(1), 311–332. https://doi.org/10.1257/aer.98.1.311.

Wilcox, C. (1992). Race, religion, region, and abortion attitudes. Sociological Analysis, 53(1), 97–105. https://doi.org/10.2307/3711464.